

point has an unlimited power supply. It attempts to lower power consumption by minimising the power wasted due to radio operation inefficiencies at each node, and thereby increases battery lifetime.

In addition, this protocol offers excellent real-time delay guarantees, congestion control and data transmission fairness. When tested on a wireless sensor network for traffic control, it demonstrated 1000 times lower power consumption than traditional contention network protocols. You can read more about this innovation at <http://techtransfer.universityofcalifornia.edu/NCD/17287.html>

## Applications in medical electronics

"One of the areas which are increasingly benefitting from wireless sensor technologies is medical electronics. According to a report by Electronics Industry Market Research and Knowledge Network, enabling wireless sensor solutions can reduce hospitalisation costs by promoting portable medical equipment. "This can collectively reduce costs for annual healthcare by \$25 billion by 2012," cites Tyagi of Freescale.

Freescale's offerings in this space



WiiMote—the key control device for Nintendo's Wii console—has fired the imagination of many developers

include the reference design for an automated home blood pressure monitor. This is an example of how the sensing, data communication and processing capabilities of Freescale's Flexis QE128 product family can interact to create a handheld medical solution. This low-cost personal BP monitor is available at affordable price-points.

The company also offers piezoresistive pressure sensors that are suitable for medical instrumentation applications like portable digital blood pressure meters that can provide key diagnostics for healthcare patients.

Another concept put forth by the company is that of a personal electrocardiogram (ECG) monitor. Tyagi explains, "The high cost of a personal ECG monitor can be reduced through the use of new technologies which incorporate a microcontroller and digital signal processing into one device. A low-cost, single-chip solution by Freescale can help overcome this obstacle."

## Smart meters to help reduce electricity bills

Rising electricity bills can shock just about anybody. Companies want to reduce their energy costs, just as consumers seek more efficient means to manage their own energy consumption. Smart meters could come to the rescue. These are essentially intelligent meters that can sense a home or organisation's power consumption in more detail than a normal meter. This information can either be used by a consumer to better understand and manage his usage or relayed back to the service provider for billing purposes.

"ZigBee-enabled meters can communicate and control ZigBee-enabled devices in the home, such as the heating, cooling and air-conditioning systems, to enable utility company programs such as load control, demand response and time-of-use pricing. These programs not only reduce the peak load on the utility grid but also help the home owners make smart decisions about their energy usage," says Tyagi. Though it might be a while

before we see this in India, it is very useful for international service providers who have variable rates for power usage based on the time of the day. Consumption during night or other 'happy hours' will be charged less than usual timings; plus consumption by certain devices might be charged more than the others.

## The planet's central nervous system

Sensor technology is so hot today that almost every IT and electronics major has some or the other related project. HP Labs, for instance, announced a very ambitious project late last year. Their dream is to build a 'central nervous system for the earth' (CeNSE) that could well be a veritable backbone for the Internet of Things that has been spoken about for so long.

Their goal is to 'make the earth speak for itself' using a network of tiny, cheap, robust and very accurate sensors that will give the computers, devices and other tech toys around us the power to sense what is going on in the environment.

Peter Hartwell, senior researcher and team lead for this project, envisions sensing nodes about the size of a pushpin stuck to bridges and buildings to warn of structural strains or weather conditions; scattered along roadsides to monitor traffic, weather and road conditions; embedded in everyday electronics to track hospital equipment, sniff out pesticides and pathogens in food; and so on. Together, the network would make life easier and safer for mankind, while also collecting valuable information that will create awareness for a better tomorrow.

The first goal towards achieving this mega dream is to develop dirt cheap sensors.

Hartwell is working on the first such sensor—an accelerometer that can detect motion and vibration. The device is sensitive enough to detect a heartbeat, and contributes a sense of 'touch and feel' to CeNSE. As per details revealed by the company, "The